

PAPER_158 $\hat{=}$ Hybrid MUGE Blending Model: $g_{\text{hybrid}} = \hat{I}^2 \hat{A} \cdot g_{\text{compressed}} + (1 - \hat{I}^2) \hat{A} \cdot g_{\text{resonance}}$

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Abstract

This paper introduces the **Hybrid MUGE Blending Model**, a continuous interpolation between the Compressed MUGE and Resonance MUGE gravity frameworks. The blending parameter $\hat{I}^2 = \exp(-B/B_{\text{crit}})$ transitions smoothly from $\hat{I}^2 \hat{=}$ 0 (pure resonance, magnetar regime) to $\hat{I}^2 \hat{=}$ 1 (pure compressed Newtonian+, weak-field regime). This unified hybrid is the **first algebraic bridge** between the Newtonian limit and the full resonance superconductive regime within the UQFF framework, extending $\hat{A}$ 2.2 PAPER_155 which proved Newtonian emergence only in the $f_{\text{TRZ}} \hat{=}$ 0 limit.

1. Motivation

PAPER_146 ($\hat{A}$ 2.2) established the 12-term resonance MUGE master equation. PAPER_090 ($\hat{A}$ 1.12) established the 9-term compressed MUGE. Previously, no **analytical bridge** existed for intermediate regimes where B is a significant fraction of B_{crit} (e.g., solar magnetosphere, white dwarfs, neutron star crusts).

The blending model provides: - Continuous function of B/B_{crit} across the full magnetic field parameter space - Smooth limiting behavior at both extremes - Single equation for all astrophysical environments

2. Hybrid Blending Equation

$$g_{\text{hybrid}}(r, t) = \beta \cdot g_{\text{comp}}(r, t) + (1 - \beta) \cdot g_{\text{res}}(r, t)$$

where

$$\beta = e^{-B/B_{\text{crit}}}$$

- g_{comp} = compressed MUGE gravity (PAPER_090, 9-term Newtonian+corrections)
 - g_{res} = resonance MUGE gravity (PAPER_146/159, 12/13-term superconductive resonance)
 - B = local magnetic field strength [T]
 - B_{crit} = critical quantum field ($4.4 \hat{\sim} 10^{11}$ T for magnetars)
-

3. Limiting Cases

3.1 Magnetar Regime ($B/B_{\text{crit}} \approx 1$)

$$\beta = e^{-1} \approx 0.368 \Rightarrow g_{\text{hybrid}} \approx 0.37 g_{\text{comp}} + 0.63 g_{\text{res}}$$

For SGR 1745-2900 ($B = 3 \times 10^{11}$ T, $B_{\text{crit}} \approx 4.4 \times 10^{13}$ T):

$$\beta_{\text{SGR}} = e^{-3 \times 10^{11} / 4.4 \times 10^{13}} \approx 0.9933 \approx 1$$

$g_{\text{hybrid}} \approx g_{\text{comp}}$ for SGR 1745 (compressed dominant near-critical but below)

3.2 Solar Regime ($B \ll B_{\text{crit}}$)

For Sun ($B_{\text{s}} = 10^{-4}$ T):

$$\beta_{\text{Sun}} = e^{-10^{-4} / 4.4 \times 10^{13}} \approx 1.000000$$

$g_{\text{hybrid}} \approx g_{\text{comp}}$ (pure Newtonian+ regime)

3.3 Neutron Star Surface ($B \sim 10^{12}$ T)

$$\beta_{\text{NS}} = e^{-10^{12} / 4.4 \times 10^{13}} \approx 0.977$$

$g_{\text{hybrid}} \approx 0.977 g_{\text{comp}} + 0.023 g_{\text{res}}$ (dominantly compressed; 2.3% resonance correction)

4. Python Implementation (CP3)

```
import math
```

```
def compute_hybrid_muge(g_compressed: float, g_resonance: float,
                        B: float, B_crit: float = 4.4e13) -> float:
    """
    Hybrid MUGE blending: g_hybrid = beta * g_comp + (1-beta) * g_res
    where beta = exp(-B/B_crit).
```

Args:

g_compressed: Compressed MUGE gravity [m/s²]
g_resonance: Resonance MUGE gravity [m/s²]
B: Local magnetic field [T]
B_crit: Critical field (default 4.4e13 T, magnetar)

Returns:

g_hybrid: Blended MUGE gravity [m/s²]

```

"""
if B_crit <= 0:
    raise ValueError("B_crit must be positive")
beta = math.exp(-B / B_crit)
return beta * g_compressed + (1.0 - beta) * g_resonance

```

5. Validation & 7-System Blending Table

System	B [T]	$\hat{I}^2$	g_{comp} [m/s $\hat{A}^2$]	g_{res} [m/s $\hat{A}^2$]	g_{hybrid} [m/s $\hat{A}^2$]
SGR 1745-2900	$3\hat{A}10\hat{A}1\hat{A}1$	0.9933	$1.783\hat{A}10\hat{A}3\hat{A}1$	$1.655\hat{A}10\hat{A}1\hat{A}1$	$1.785\hat{A}10\hat{A}1\hat{A}1$ + domin
Sagittarius A*	$1\hat{A}10\hat{A}1\hat{A}1$	1.0	$1.816\hat{A}10\hat{A}3\hat{A}1$	$1.256\hat{A}10\hat{A}1\hat{A}1$	$1.816\hat{A}10\hat{A}3\hat{A}1$ g_comp
Tapestry	$1\hat{A}10\hat{A}1\hat{A}1$	1.0	$2.989\hat{A}10\hat{A}3\hat{A}1$	$1.257\hat{A}10\hat{A}1\hat{A}1$	$2.989\hat{A}10\hat{A}3\hat{A}1$ g_comp
Westerlund 2	$1\hat{A}10\hat{A}1\hat{A}1$	1.0	$2.989\hat{A}10\hat{A}3\hat{A}1$	$1.257\hat{A}10\hat{A}1\hat{A}1$	$2.989\hat{A}10\hat{A}3\hat{A}1$ g_comp
Pillars	$1\hat{A}10\hat{A}1\hat{A}1$	1.0	$1.989\hat{A}10\hat{A}2\hat{A}1$	$1.256\hat{A}10\hat{A}1\hat{A}1$	$1.989\hat{A}10\hat{A}2\hat{A}1$ g_comp
Rings	$1\hat{A}10\hat{A}1\hat{A}1$	1.0	$2.989\hat{A}10\hat{A}3\hat{A}1$	$1.257\hat{A}10\hat{A}1\hat{A}1$	$2.989\hat{A}10\hat{A}3\hat{A}1$ g_comp
Student's Guide	$1\hat{A}10\hat{A}1\hat{A}1$	0	$2.000\hat{A}10\hat{A}1\hat{A}1$	$1.257\hat{A}10\hat{A}1\hat{A}1$	$2.000\hat{A}10\hat{A}1\hat{A}1$ g_comp

Note: For all non-magnetar systems, $\hat{I}^2 \hat{A}1$ so $g_{\text{hybrid}} \hat{A} g_{\text{comp}}$. The blending becomes significant only for systems with $B/B_{\text{crit}} > 0.01$ (i.e., $B > 4.4\hat{A}10\hat{A}1\hat{A}1$ T).

6. Theoretical Significance

The hybrid model provides: 1. **A unified equation** replacing case-by-case selection of compressed vs. resonance 2. **Continuous parameter space** enabling interpolation between observational regimes 3. **Automatic mode selection**: $\hat{I}^2$ encodes the magnetic suppression factor from PAPER_090 4. **Extends PAPER_155**: While PAPER_155 proves Newtonian emergence at $f_{\text{TRZ}} \hat{A}0$, this model handles intermediate B fields without requiring $f_{\text{TRZ}} \hat{A}0$

Connection to 4 UQFF Operational Modes (PAPER_064): - Compressed: $\hat{I}^2 = 1$ - Buoyant: $\hat{I}^2$ and Ubi terms active - Resonant: $\hat{I}^2 = 0$ - Superconductive: $\hat{I}^2 = 0$, f_{TRZ} active

Status: $\hat{A}$ Complete | **CP Stage:** CP3 (new compute_hybrid_muge() function) **Supersedes:** N/A (new model) | **Related:** PAPER_090 (compressed), PAPER_146 (12-term resonance), PAPER_155 (Newtonian limit), PAPER_064 (4 modes)

UQFF computed: MUGE buoyancy ratio $U_{\text{bi}}/F_{\text{U}} = [\text{SSq}] \times ? \times r^2/\text{GM} = 5.7\text{e-}1 \times 5.0\text{e-}4 = 2.85\text{e-}4$; compressed MUGE baseline $g = 5.4\text{e-}7$ m/s $\hat{A}^2$ at r_{ISCO} .